

Outcomes for posterior uveal melanoma: Validation of American Brachytherapy Society Guidelines

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ABSTRACT

PURPOSE: To assess outcomes of small and medium choroidal melanoma (less than 5.0 mm in height) following Iodine-125 episcleral brachytherapy.

METHODS AND MATERIALS: Patients with small and medium choroidal melanoma that underwent Iodine-125 brachytherapy with apical height of 1.0 mm to 5.0 mm and largest basal diameter of ≤ 16.0 mm were included. Data were extracted from the original dosimetry plans to determine doses to vision critical structures with the prescription point to the apical height (actual dose, ABS guidelines) and, after simulation, with the prescription point to the height of 5.0 mm (simulated dose, COMS protocol). Visual acuity (VA) outcomes with actual dose and that predicted with the simulated dose were estimated along with local recurrence, ocular survival, and survival at 5 years.

RESULTS: A total of 339 patients with a mean age of 61.5 years with a mean follow up duration of 43.4 months were included. The mean dose reduction for lens, optic disc, and fovea was 34%, 39.4%, and 41.4%, respectively with actual dose when compared with simulated dose. The Kaplan-Meier estimations for 3 year event free rate of VA of 20/50 or better were 56% and 31% for actual dose and simulated dose, respectively. Only 3 events of local recurrence were observed (enucleated) yielding 5 year local control and ocular survival rate of 98%. Overall survival (OS) and metastasis free survival (MFS) were 95% and 87.5% at 5 years, respectively.

CONCLUSIONS: Small and medium choroidal melanoma treated according to ABS has excellent outcomes. Brachytherapy planning using ABS guidelines as compared to COMS protocol may be associated with lower rates of radiation toxicity and vision loss. © 2021 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

ABS guidelines; COMS protocol; Choroidal melanoma; Brachytherapy; Iodine-125; Vision loss

Introduction

Brachytherapy is the most frequent primary treatment for uveal melanoma in the United States (1). A recent survey has identified a variation in the brachytherapy planning of which therapeutic dose and prescription point are

most relevant (2). Therapeutic dose for brachytherapy using Iodine-125 of 85 Gy as prescribed in the Collaborative Ocular Melanoma Study (COMS) has become the standard (3). However, the prescription point was variable depending upon the tumor height; for tumors more than 5.0 mm in height, the prescription point was the actual tumor height, whereas for tumors less than 5.0 mm in height, the prescription point was adjusted to 5.0 mm. In 2012, the American Brachytherapy Society (ABS) (TG-129 report) issued a recommendation of using either actual tumor height as the prescription point or 5.0 mm for tumors less than 5 mm (4). The guidelines were revised by ABS in 2014 (Level 1 consensus) specifying that the prescription point should be the actual tumor height for all tumors including those that were less than 5.0 in height (5). The panel cited concerns

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